Amazon Fresh Analysis Documentation

PROJECT DOCUMENTATION

Design and Analysis of an E-Commerce Database Using MySQL

Domain: E-Commerce and Retail Analysis

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Domain	E-Commerce and Retail Analytics
Dataset	Amazon Fresh Analytics
Tools Used	MYSQL Workbench 8.0

1. Project Overview

Objective: The primary objective of this project is to design and implement a relational database for Amazon Fresh using MySQL. The project focuses on organizing customer, product, supplier, order, and review data into a structured database that supports efficient data management and business analysis.

The project demonstrates how SQL can be used to store, retrieve, manipulate, and analyse business data. Various SQL concepts such as DDL, DML, Joins, Aggregations, Subqueries, Constraints, Normalization, and ACID Transactions are implemented to solve real-world business problems.

Business Need:

Amazon Fresh is an online grocery and daily essentials platform that manages thousands of customer transactions every day. As business operations expand, handling large amounts of data efficiently becomes increasingly important.

Expected Outcome:

- After completing this project, the database can:
- Store business data securely.
- Reduce redundancy through normalization.
- Improve data consistency.
- Generate analytical reports.
- Support business decision-making using SQL.

2. Dataset Description

The Amazon Fresh Analytics database is designed to manage and analyze data related to customers, products, suppliers, orders, and customer reviews. The dataset consists of six interconnected tables that support efficient storage, retrieval, and analysis of business information.

Dataset Tables

Customers: Stores customer information including personal details, city, age, and Prime membership status.

Products: Stores product details such as product name, category, price, stock quantity, and supplier information.

Suppliers: Contains supplier details including supplier name, city, and contact number.

Orders: Stores order information such as customer ID, order date, and total order amount.

Order-Details: Contains detailed information about products purchased in each order including quantity and subtotal.

Reviews: Stores customer ratings and reviews for purchased products.

3. Data Base Design

A relational database model was implemented using MySQL Workbench. The database consists of six related tables connected through primary keys and foreign keys.

The design follows relational database principles to minimize redundancy and maintain data integrity.

Database Features

Relational Database Structure

Primary Keys

Foreign Keys

Referential Integrity

Normalized Tables

Efficient Query Performance

4. ER-Diagram Explanation

An Entity Relationship (ER) Diagram was created using MySQL Workbench to visualize the database structure and relationships.

The ER Diagram includes six entities:

- Customers
- Products
- Suppliers
- Orders
- Order-Details
- Reviews

Benefits of ER Diagram:

- Provides a clear visualization of the database.
- Identifies relationships between tables.
- Reduces data redundancy.

5. SQL Constraints

SQL Constraints were applied to maintain data integrity and improve database reliability.

Primary Key:

A Primary Key uniquely identifies every record in a table.

Foreign Keys establish relationships between parent and child tables.

Examples:

- Orders → Customers
- Products → Suppliers
- Order-Details → Orders
- Order-Details → Products
- Reviews → Customers
- Reviews → Products

UNIQUE Constraint

The UNIQUE constraint prevents duplicate values.

Example:

Customer Name was defined as UNIQUE.

CHECK Constraint

The CHECK constraint validates data before insertion.

Examples:

Age > 18

Rating BETWEEN 1 AND 5

Quantity > 0

DEFAULT Constraint

The DEFAULT constraint automatically inserts a predefined value.

Example:

Prime Member = 'No'

NOT NULL Constraint

NOT NULL ensures mandatory fields cannot be left empty.

Applied to:

- Customer Name
- Age
- Product Name
- Price
- Order Date
- Contact Number

6. Data Manipulation Language (DML)

Overview

Data Manipulation Language (DML) is used to insert, update, and delete records from the database.

The following operations were completed as part of the project.

INSERT Operation (Task 5)

Three new product records were inserted into the Products table.

Purpose

- Add new products.
- Expand inventory.
- Test database functionality.

UPDATE Operation (Task 6)

The stock quantity of a selected product was updated using the ProductID.

Purpose

- Maintain accurate inventory.
- Reflect stock changes after purchases or restocking.

DELETE Operation (Task 7)

A supplier record was deleted based on a specified city.

Purpose

- Remove obsolete supplier records.
- Maintain an updated supplier database.
- Benefits of DML Operations
- Supports day-to-day database management.
- Keeps records current and accurate.
- Enables efficient transaction processing.
- Improves data quality.

7. SQL Analysis (Task 1-14)

Overview

The SQL analysis phase focused on answering real-world business questions using SQL queries. Various SQL concepts such as filtering, joins, aggregate functions, grouping, sorting, subqueries, and ranking techniques were used to generate meaningful business insights from the Amazon Fresh database.

The analysis helped identify customer purchasing behaviour, product demand, supplier performance, sales trends, and Prime membership distribution.

Task 1 – ER Diagram

Objective

Create an Entity Relationship (ER) Diagram to understand how all database tables are connected.

Work Performed

An ER Diagram was created using MySQL Workbench showing:

- Customers
- Products
- Suppliers
- Orders
- Order-Details
- Reviews
- Outcome
- Clear visualization of relationships.
- Better understanding of database structure.
- Reduced chances of database design errors.

Task 2 – Primary Key & Foreign Key Relationships

Objective

Identify all Primary Keys and Foreign Keys.

Result

Primary Keys uniquely identify each record, while Foreign Keys establish relationships between related tables.

- Business Benefit
- Maintains data integrity.
- Prevents duplicate records.
- Supports efficient joins.

Task 3 – Basic SQL Queries

Objective

Retrieve business information using simple SQL queries.

- Queries Performed
- Retrieve customers from a specific city.
- Retrieve products under the Fruits category.
- SQL Concepts Used
- SELECT
- FROM
- WHERE

Outcome

Basic filtering operations successfully retrieved the required records.

Task 4 – DDL Statements

Objective

Recreate the Customers table with constraints.

- Constraints Applied
- Primary Key
- NOT NULL
- CHECK
- UNIQUE
- DEFAULT

Outcome

A properly structured Customers table was created with all required validations.

Task 5 – Insert Records

Objective

Insert three new products into the Products table.

Outcome

- New products were successfully added without affecting existing records.
- Business Benefit
- Supports continuous inventory expansion.

Task 6 – Update Records

Objective

Update stock quantity using Product-ID.

Outcome

- Inventory values were updated successfully
- Business Benefit
- Maintains accurate stock information.

Task 7 – Delete Records

Objective

Delete supplier information based on city.

Outcome

- Selected supplier records were removed successfully.
- Business Benefit
- Removes inactive or unnecessary supplier records.

Task 8 – SQL Constraints

Objective

Apply additional constraints.

- Constraints Used
- CHECK (Rating 1–5)
- DEFAULT (Prime Member = No)

Outcome

Improved data quality and prevented invalid entries.

Task 9 – Clauses & Aggregations

WHERE Clause

Used to filter orders placed after 01-Jan-2024.

GROUP BY

Used to summarize sales data.

HAVING

Displayed products having an average rating greater than 4.

ORDER BY

Ranked products based on total sales.

Aggregate Functions Used

- SUM()
- AVG()
- COUNT()
- MAX()
- MIN()

Business Benefit

Provides summarized information for management reporting.

Task 10 – High Value Customers

Objective

Identify customers who spent more than ₹5,000.

SQL Concepts Used

- SUM()
- GROUP BY
- HAVING

Analysis

Each customer's total spending was calculated and ranked.

Business Insight

High-value customers contribute significantly to overall revenue and can be targeted through loyalty programs and exclusive offers.

Task 11 – Complex Joins

Objective

Analyse revenue and supplier performance.

- Joins Used
- INNER JOIN
- Multiple Table JOIN
- Analysis Performed
- Total revenue per order.
- Customers placing the highest number of orders.
- Supplier with maximum products in stock.

Business Benefit

Provides detailed insights into customer behaviour and supplier efficiency.

Task 12 – Database Normalization

Objective

Normalize Products table into Third Normal Form (3NF).

- Steps
- Separate Categories into a new table.
- Separate Subcategories.
- Create Foreign Key relationships.
- Benefits
- Reduced redundancy.
- Improved consistency.
- Easier maintenance.
- Better scalability.

Task 13 – Subqueries

Objective

Solve business problems using nested queries.

- Analysis Performed
- Top 3 products based on revenue.
- Customers who have not placed any orders.
- SQL Concepts Used
- Nested SELECT
- IN
- NOT IN
- LIMIT

Business Benefit

Identifies top-performing products and inactive customers.

Task 14 – Business Insights

Objective

Generate actionable business reports.

Analysis Performed

- Cities with the highest concentration of Prime Members.
- Top 3 most frequently ordered product categories.

Business Benefit

Supports marketing strategies, inventory planning, and customer engagement.

8. Project Deliverable

The following deliverables were successfully completed during the project.

- Amazon Fresh Relational Database
- ER Diagram
- SQL Scripts
- DDL Statements
- DML Statements
- Constraints Implementation
- SQL Queries (Task 1–14)
- Normalized Database (3NF)
- Business Analysis Reports
- Project Documentation

9. Result

The Amazon Fresh Analytics project successfully achieved all the planned objectives.

Key Achievements:

- Designed a normalized relational database.
- Created an ER Diagram showing table relationships.
- Applied SQL constraints to maintain data integrity.
- Executed DDL and DML operations successfully.
- Performed analytical SQL queries for business reporting.
- Generated meaningful customer, product, supplier, and revenue insights.
- Applied ACID principles to improve transaction reliability.

11. Conclusion

The Amazon Fresh Analytics project successfully demonstrated the practical implementation of SQL and relational database concepts in an e-commerce environment.

The database was designed efficiently using normalization techniques and appropriate relationships between entities. SQL queries were developed to answer important business questions related to customers, products, suppliers, orders, and reviews.

The project showcased the effective use of DDL, DML, joins, aggregate functions, subqueries, constraints, normalization, and ACID transactions. The generated business insights can support decision-making, improve inventory management, enhance customer satisfaction, and optimize overall business operations.

Overall, this project strengthened practical knowledge of database design and SQL while demonstrating how relational databases can solve real-world business problems effectively.